

Ex. 8.8

Ques: 1

$$\int_0^{\infty} \frac{1}{x^2+1} dx$$

$$= \lim_{b \rightarrow \infty} \int_0^b \frac{1}{x^2+1} dx$$

$$= \lim_{b \rightarrow \infty} \left| \tan^{-1} x \right|_0^b$$

$$= \lim_{b \rightarrow \infty} (\tan^{-1} b - \tan^{-1} 0)$$

$$= \lim_{b \rightarrow \infty} (\tan^{-1} b - 0)$$

$$= \tan^{-1} \infty$$

$$= \frac{\pi}{2}$$

Ques: 9 Ex. 8.8

$$\int_{-\infty}^{\infty} \frac{2}{x^2-1} dx$$

$$\frac{1}{x^2-1} = \frac{1}{(x-1)(x+1)} = \frac{A}{(x-1)} + \frac{B}{(x+1)}$$

$$1 = A(x+1) + B(x-1)$$

$$\text{Let } x = -1$$

$$1 = A(-1+1) + B(-1-1)$$

$$1 = -2B$$

$$B = -\frac{1}{2}$$

$$1 = A(x+1) + B(x-1)$$

$$\text{Let } x = 1$$

$$1 = A(1+1) + B(1-1)$$

$$1 = 2A$$

$$A = \frac{1}{2}$$

$$\frac{1}{x^2-1} = \frac{1/2}{(x-1)} + \frac{-1/2}{(x+1)}$$

$$= \frac{1}{2(x-1)} - \frac{1}{2(x+1)}$$

Now $\int_{-\infty}^{-2} \frac{2}{x^2-1} dx$

$$= 2 \int_{-\infty}^{-2} \frac{1}{x^2-1} dx$$

$$= 2 \int_{-\infty}^{-2} \left[\frac{1}{2(x-1)} - \frac{1}{2(x+1)} \right] dx$$

$$= \int_{-\infty}^{-2} \frac{1}{x-1} dx - \int_{-\infty}^{-2} \frac{1}{x+1} dx$$

$$= \lim_{a \rightarrow -\infty} \int_a^{-2} \frac{1}{x-1} dx - \lim_{a \rightarrow -\infty} \int_a^{-2} \frac{1}{x+1} dx$$

$$= \lim_{a \rightarrow -\infty} \left| \ln|x-1| \right|_a^{-2} - \lim_{a \rightarrow -\infty} \left| \ln|x+1| \right|_a^{-2}$$

$$= \lim_{a \rightarrow -\infty} \left| \frac{\ln|x-1|}{\ln|x+1|} \right|_a^{-2}$$

$$\left[\begin{array}{l} \ln a - \ln b = \ln\left(\frac{a}{b}\right) \\ \ln a + \ln b = \ln(ab) \end{array} \right]$$

$$= \lim_{a \rightarrow -\infty} \left| \frac{\ln |-3|}{\ln |-1|} - \frac{\ln |a-1|}{\ln |a+1|} \right|$$

$$= \left[\ln |-3| - \ln |-1| \right] - \lim_{a \rightarrow -\infty} \ln \left| \frac{a-1}{a+1} \right|$$

$$= \ln 3 - 0 - \lim_{a \rightarrow -\infty} \ln \left| \frac{a-1}{a+1} \right|$$

$$= \ln 3 - \ln \left[\lim_{a \rightarrow -\infty} \frac{\frac{a-1}{a}}{\frac{a+1}{a}} \right]$$

$$= \ln 3 - \ln \left[\frac{1 - \lim_{a \rightarrow -\infty} \frac{1}{a}}{1 + \lim_{a \rightarrow -\infty} \frac{1}{a}} \right]$$

$$= \ln 3 - \ln \left[1 - \frac{\frac{1}{-\infty}}{\frac{1}{-\infty}} \right] = \ln 3 - \ln 1 = \ln 3$$